WORKED SOLUTION

STEP II 2025, Question 11

Probability and Statistics • Expectation and Binomial Sums

STEP II 2025, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Both games end at the first $2k$, so each is a sum over the number of rolls r. For Ali the sum is geometric-like, and part (i) is exactly the tool for the expected number of rolls. For Zen, a winning game is fixed by *which* numbers appear before the $2k$, because once you know the set, the increasing order is forced. Counting these sets brings in binomial coefficients, so the sum is the one from part (iii).

Solution

Part (i)

For $|x| < 1$ the geometric series gives

$$\sum_{r=0}^{\infty} x^r = \frac{1}{1-x}.$$

A power series can be differentiated term by term inside its interval of convergence, so differentiating both sides with respect to x gives

$$\sum_{r=1}^{\infty} r x^{r-1} = \frac{1}{(1-x)^2}, \qquad |x| < 1,$$

as required.

Part (ii)

On any one roll there are three kinds of outcome:

$$\mathrm{P}(\text{a } 2k) = \frac{1}{2k}, \qquad \mathrm{P}(\text{even, not } 2k) = \frac{k-1}{2k}, \qquad \mathrm{P}(\text{odd}) = \frac{1}{2}.$$

Let R be the number of rolls. Ali wins with $R = r$ exactly when rolls $1, \ldots, r-1$ are even numbers other than $2k$ and roll r is $2k$. The rolls are independent, so with $x = \dfrac{k-1}{2k}$,

$$\mathrm{P}(\text{Ali wins and } R = r) = \frac{1}{2k}\, x^{r-1}.$$

(a) Summing the geometric series (note $0 \le x < 1$):

$$\mathrm{P}(\text{Ali wins}) = \frac{1}{2k} \cdot \frac{1}{1-x} = \frac{1}{2k} \cdot \frac{2k}{k+1} = \frac{1}{k+1}.$$

There is a quick way to see this. Ignore the even numbers other than $2k$, since they don't affect the result. The game is decided by whichever comes first, an odd number or the $2k$. There are k odd faces and one face showing $2k$, all equally likely, so the chance that the $2k$ comes first is $\frac{1}{k+1}$.

(b) Ali earns £r if he wins with $R = r$ and nothing otherwise, so by part (i)

$$\mathrm{E}(\text{Ali's earnings}) = \sum_{r=1}^{\infty} r \cdot \frac{1}{2k}\, x^{r-1} = \frac{1}{2k} \cdot \frac{1}{(1-x)^2} = \frac{1}{2k}\left(\frac{2k}{k+1}\right)^2 = \frac{2k}{(k+1)^2} \text{ pounds.}$$

Part (iii)

Call the expression S. Its general term is $(r+1)\binom{n}{r}x^r$ for $r = 0, 1, \ldots, n$. The factor $r+1$ suggests differentiating x^{r+1}:

$$S = \sum_{r=0}^{n} (r+1)\binom{n}{r}x^r = \frac{\mathrm{d}}{\mathrm{d}x}\left(\sum_{r=0}^{n}\binom{n}{r}x^{r+1}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\Big(x(1+x)^n\Big),$$

using the binomial theorem. By the product rule,

$$S = (1+x)^n + nx(1+x)^{n-1} = (1+x)^{n-1}(1+(n+1)x).$$

Part (iv)

Suppose Zen wins with exactly r rolls. The last roll is $2k$, and the first $r-1$ rolls are strictly increasing numbers from $\{1, 2, \ldots, 2k-1\}$. (They must all be less than $2k$ anyway, since the game stops at the first $2k$.)

> **Ask yourself** How many winning sequences of length r are there?

A strictly increasing sequence is fixed once we know which numbers appear in it, because there is only one way to put a set of distinct numbers in increasing order. So the winning sequences of length r correspond to the $(r-1)$-element subsets of $\{1, \ldots, 2k-1\}$. There are $\binom{2k-1}{r-1}$ of them, and $1 \le r \le 2k$.

Each particular sequence of r rolls has probability $\left(\frac{1}{2k}\right)^r$. Hence

$$\mathrm{E}(\text{Zen's earnings}) = \sum_{r=1}^{2k} r\binom{2k-1}{r-1}\left(\frac{1}{2k}\right)^r = \frac{1}{2k}\sum_{j=0}^{2k-1}(j+1)\binom{2k-1}{j}\left(\frac{1}{2k}\right)^j,$$

where $j = r - 1$. This is the sum in part (iii) with $n = 2k - 1$ and $x = \frac{1}{2k}$, and then $1 + (n+1)x = 1 + 2k \cdot \frac{1}{2k} = 2$. Therefore

$$\mathrm{E}(\text{Zen's earnings}) = \frac{1}{2k}\left(1 + \frac{1}{2k}\right)^{2k-2} \cdot 2 = \frac{1}{k}\left(1 + \frac{1}{2k}\right)^{2k-2} \text{ pounds.}$$

Part (v)

Divide Zen's expected earnings by Ali's:

$$\rho = \frac{1}{k}\left(1 + \frac{1}{2k}\right)^{2k-2} \cdot \frac{(k+1)^2}{2k} = \frac{(k+1)^2}{2k^2} \cdot \frac{\left(1 + \frac{1}{2k}\right)^{2k}}{\left(1 + \frac{1}{2k}\right)^{2}}.$$

Now let k be large. Then

- $\dfrac{(k+1)^2}{k^2} = \left(1 + \dfrac{1}{k}\right)^2 \approx 1$;
- $\left(1 + \dfrac{1}{2k}\right)^2 \approx 1$;
- $\left(1 + \dfrac{1}{2k}\right)^{2k} \approx \mathrm{e}$, by the given approximation with $n = 2k$.

So

$$\rho \approx \frac{\mathrm{e}}{2} \approx \frac{2.71828}{2} = 1.359\ldots,$$

and for large k Zen's expected earnings are about 1.359 times Ali's, which is a little over 35% more, as required.

Checks

- For $k = 1$ the die is a coin with faces $1, 2$. Ali wins only if the first roll is 2, which has probability $\frac{1}{2} = \frac{1}{k+1}$, and then earns £1, giving expected earnings $\frac{1}{2} = \frac{2k}{(k+1)^2}$. Zen wins with the sequences 2 (probability $\frac{1}{2}$, earning £1) and $1, 2$ (probability $\frac{1}{4}$, earning £2), so her expected earnings are $\frac{1}{2} + \frac{1}{2} = 1$. The formula agrees: with $k = 1$ the exponent $2k - 2$ is 0, so it gives $\frac{1}{1} \cdot 1 = 1$.
- For $k = 3$, Ali's expected earnings are $\frac{3}{8}$ and Zen's are $\frac{1}{3}\left(\frac{7}{6}\right)^4 = \frac{2401}{3888} \approx 0.618$. Both agree with a direct enumeration of all winning sequences and with a computer simulation of the games.
- The ratio ρ is about 1.456 at $k = 10$, 1.369 at $k = 100$ and 1.3592 at $k = 10^4$. It is decreasing towards $\mathrm{e}/2 = 1.35914\ldots$.

Common slips

- In part (iv), counting ordered sequences with a factorial instead of subsets. There is only one increasing arrangement of each set.
- Forgetting that the earlier rolls in Zen's game can't include $2k$. This makes the range $\{1, \ldots, 2k-1\}$, so $n = 2k-1$, not $2k$.
- In part (ii), using $\frac{k}{2k}$ instead of $\frac{k-1}{2k}$ as the probability of continuing. The roll $2k$ is even, but it ends the game.

Beyond STEP

Part (iii) is a small example of *generating functions*: you encode a sequence as the coefficients of a power series and then use calculus on the series. Probability generating functions take this further. If $G(s) = \mathrm{E}(s^X)$, then $\mathrm{E}(X) = G'(1)$, and that is exactly the differentiation trick used twice here. The limit $\left(1 + \frac{1}{n}\right)^n \to \mathrm{e}$ turns up all over probability, for instance in the Poisson approximation to the binomial. Grimmett and Welsh, *Probability: An Introduction*, has a good chapter on generating functions.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.